Generating Individual-Level Full-Day Activity Trajectory Dataset via Large Language Models

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Abstract

Human mobility data are essential for a wide range of applications, yet their use is constrained by privacy concerns and data utility. Synthetic mobility data provide alternatives to raw mobility data, thereby reducing reliance on directly observed data while preserving analytical value. To this end, we propose an LLM-based framework for conditional individual-level daily activity trajectory generation. The framework models trajectory generation as driven by habitual patterns and contextual motivations. It introduces two key innovations. First, we establish a data-driven pipeline for persona construction, where personas define behavioural profiles of individuals to guide realistic and diverse trajectory generation, improving method reproducibility and transferability across datasets. Second, we employ an enhanced travel summary that incorporates both primary and secondary activities to define trajectory structure, enabling structurally complete trajectory generation. Experiments on a Greater London dataset demonstrate that the proposed framework achieves higher fidelity to observed mobility characteristics than representative baselines.
Keywords: human mobility, synthetic mobility data, large language models

1. Introduction

Human mobility data capture the movements of individuals or groups across space and time (Barbosa et al. 2018). They provide an essential empirical basis for understanding human movement (Song et al. 2010) and are crucial for a wide range of applications, such as commuting pattern analysis (Kung et al. 2014), epidemic modelling (Tizzoni et al. 2014), social interaction analysis (Toole et al. 2015), and many more. However, the use of human mobility data, particularly at the individual level, presents challenges in privacy and utility. High-resolution data, such as GPS data and location-based service (LBS) data, contain sensitive personal information, raising significant privacy concerns (K. Zhao et al. 2016; H. Jiang et al. 2022). In contrast, more controlled data, such as travel surveys, often suffer from limited sample sizes and self-reporting errors (Stopher and Greaves 2007), reducing their analytical value. These limitations have motivated the develop-

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ment of synthetic mobility data (Berke et al. 2022; Zhu, Ye, Wu, et al. 2023), which aim to provide alternatives to raw mobility data while preserving analytical value by learning and reproducing key statistical and behavioural patterns from real-world data (Jordon et al. 2022).

Generating synthetic mobility data remains challenging due to the complex spatiotemporal characteristics of human movement (X. Wang et al. 2023). Early statistical models struggle to capture long-range dependencies (Kulkarni et al. 2019), while deep learning models often lack interpretability (Rudin 2019). Recently, large language models (LLMs) have emerged as a new approach, with applications ranging from individual-level next-location prediction to activity sequence generation (X. Wang et al. 2023; J. Wang et al. 2024). Despite this progress, adapting general-purpose LLMs to synthetic mobility data generation remains non-trivial, and existing studies remain limited in two important respects. First, they rarely explicitly distinguish between primary (e.g., home and work) and secondary (e.g., eating, shopping, and other discretionary) activities, and are therefore not specifically designed to generate structurally complete daily trajectories. Second, when personas (i.e., prompt-based descriptions of individuals' roles and characteristics) are used to guide realistic and diverse trajectory generation, they are typically constructed manually, limiting reproducibility and transferability across datasets.

To address these limitations, we propose an LLM-based framework for conditional individual-level daily activity trajectory generation that produces structurally complete, realistic, and diverse trajectories. The framework models trajectory generation as a process jointly determined by habitual patterns and contextual motivations, where habitual patterns capture stable behavioural regularities and contextual motivations introduce short-term, event-driven variation. To ensure structural completeness, diversity, and realism, we condition the generation process through two prompt-based components: a travel summary, which defines the structure of the generated trajectory, and personas, which serve as semantic priors to guide realistic and diverse generation. Our innovations lie in these two components. First, we propose an enhanced travel summary that explicitly incorporates both primary and secondary activities, enabling the generation of structurally complete daily trajectories. Second, we develop a data-driven persona construction pipeline, in which personas are automatically derived from real-world data rather than manually specified, improving reproducibility and enabling transferability across datasets.

To validate the proposed framework, we conduct experiments on a mobility dataset in the Greater London area and compare our approach with representative baseline methods. The results demonstrate that our framework achieves higher fidelity to real-world mobility data. Ablation studies further show that both the enhanced travel summary and the data-driven persona construction contribute substantially to the observed performance improvements.

The contributions of this paper are as follows:

- We propose an LLM-based framework for conditional individual-level daily activity trajectory generation, enabling high-fidelity, synthetic alternatives to raw mobility trajectories.
- We introduce an enhanced travel summary that explicitly models primary and secondary activities, enabling the generation of structurally complete trajectories.

- We develop a data-driven persona construction pipeline that enables reproducible and transferable persona generation across datasets.
- We demonstrate through experiments on real-world data that the proposed framework achieves higher fidelity to real-world mobility characteristics than existing methods.

2. Related Work

2.1. Synthetic Mobility Data Generation

Synthetic mobility generation builds on the broader literature on mobility modelling and trajectory prediction. Empirical studies have shown that human mobility exhibits strong regularity and predictability, providing a theoretical foundation for modelling and generating mobility data (González, Hidalgo, and Barabási 2008; Song et al. 2010). Early approaches to synthetic mobility generation primarily relied on statistical models, particularly Markov-based methods, which estimate transition probabilities between locations from historical trajectories (Ashbrook and Starner 2002; Gambs, Killijian, and Del Prado Cortez 2012; Mathew, Raposo, and Martins 2012; Qiao et al. 2018; H. Wang et al. 2021). While these models are interpretable, they are constrained by fixed-order assumptions and fail to capture long-range spatiotemporal dependencies (Matamalas, De Domenico, and Arenas 2016; Kulkarni et al. 2019).

With the development of deep learning, these models have been introduced to better capture complex sequential patterns. Recurrent neural networks (RNNs) and their variants, including LSTM and GRU, improve sequence modelling compared to statistical approaches (Liu et al. 2016; Krishna et al. 2018; Feng et al. 2018). More recent work has explored a range of deep learning approaches, including generative adversarial networks (GANs), graph neural networks (GNNs), transformer-based models, and diffusion-based generative models (Yu et al. 2017; Lim et al. 2020; Xue et al. 2021; Zhu, Ye, Zhang, et al. 2023). However, despite increasing model complexity, performance gains remain limited (X. Wang et al. 2023), and these approaches generally lack interpretability (Rudin 2019).

2.2. LLM-based Human Behaviour and Mobility Modelling

LLMs are large-scale Transformer-based architectures trained on massive textual datasets (W. X. Zhao et al. 2023). They have recently been adopted for modelling complex human behaviours, owing to their capabilities in semantic understanding and in-context learning, which support nuanced reasoning about real-world scenarios and enable models to infer tasks directly from prompts (Dong et al. 2023; J. Wang et al. 2024). Building on these properties, recent work has explored LLM-based autonomous agents that integrate mechanisms such as memory, planning, and reflection to simulate coherent behavioural sequences and emergent dynamics (Park et al. 2023). Such approaches have been applied across domains, including economic modelling (N. Li et al. 2024), epidemic simulation (Williams et al. 2023), and user behaviour modelling in recommendation systems (L. Wang et al. 2025), demonstrating the potential of LLMs as a promising framework for behaviour simulation.

In the domain of human mobility modelling, LLMs have been applied to next-location prediction (X. Wang et al. 2023) and subsequently extended to the generation of sequences of daily activities (J. Wang et al. 2024). To improve the diversity and realism of generated outputs, personas—descriptions of individuals' roles and characteristics specified in prompts—have been introduced as conditioning mechanisms (X. Li et al. 2024; J. Wang et al. 2024). However, existing studies provide limited support for the generation of structurally complete trajectories and often lack a systematic, data-driven process for persona construction. This work addresses these gaps by enabling structurally complete trajectory generation and introducing a data-driven persona construction approach (*Figure 1*).

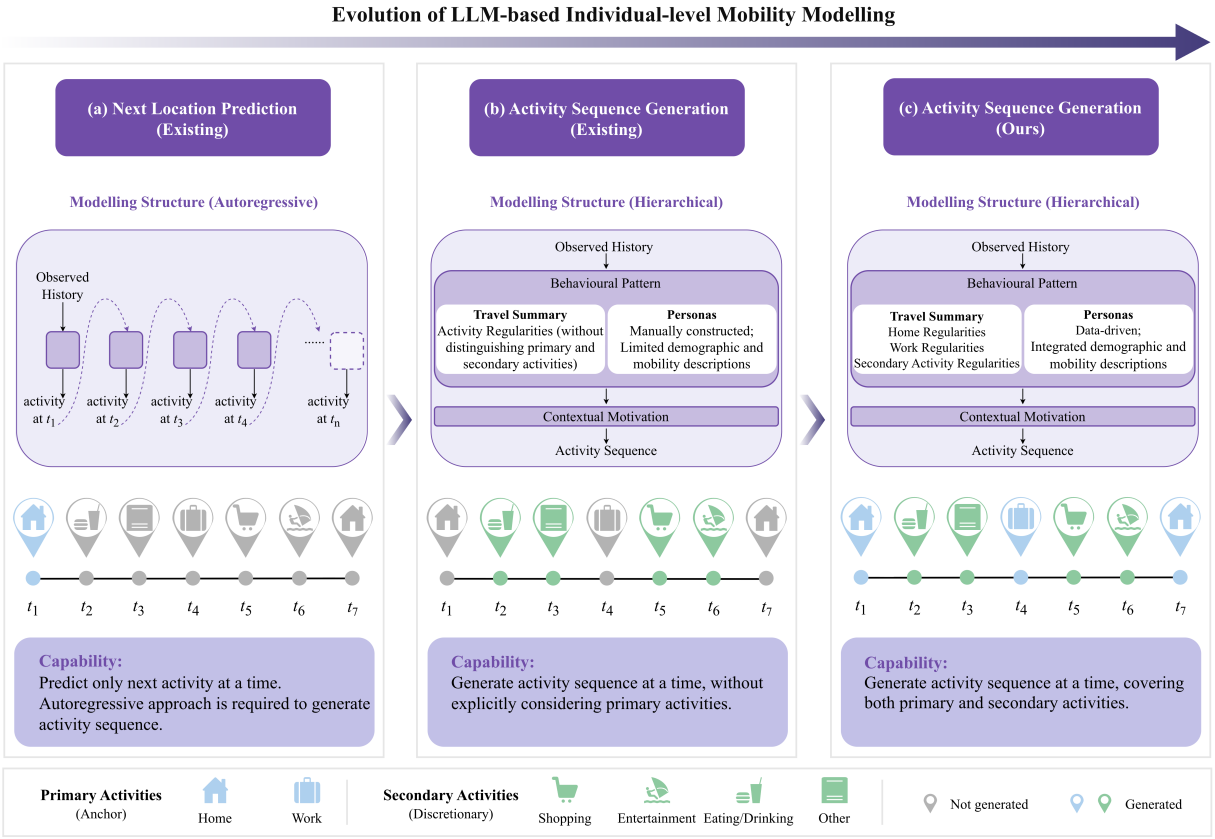


Figure 1: Evolution of individual-level mobility modelling using LLMs.

3. Methodology

3.1. Preliminaries

We formulate the task as conditional individual-level daily activity trajectory generation. Using activity records derived from the activity–trip chains of Zhong et al. (2024), an activity is defined as a stay at a location labelled with a specific travel purpose and temporal information. A trajectory is a time-ordered sequence of such activities performed by an individual within a day. Given an individual's historical trajectories over the past n days, the goal is to generate the trajectory for the target day conditioned on this history.

3.2. Framework Overview

Figure 2 shows the overall framework. We adopt a trajectory generation mechanism that models an individual's daily activity trajectory as driven by the interaction between habitual patterns and contextual motivations (J. Wang et al. 2024). Habitual patterns capture long-term behavioural regularities, reflecting stable routines such as commuting structures and frequently visited locations. In contrast, contextual motivations represent short-term, situation-specific variations driven by factors such as weather conditions, social engagements, or unexpected events (X. Wang et al. 2023). This pattern–motivation pair forms the core mechanism for trajectory generation and has proven effective in human mobility modelling (S. Jiang et al. 2016; Feng et al. 2018).

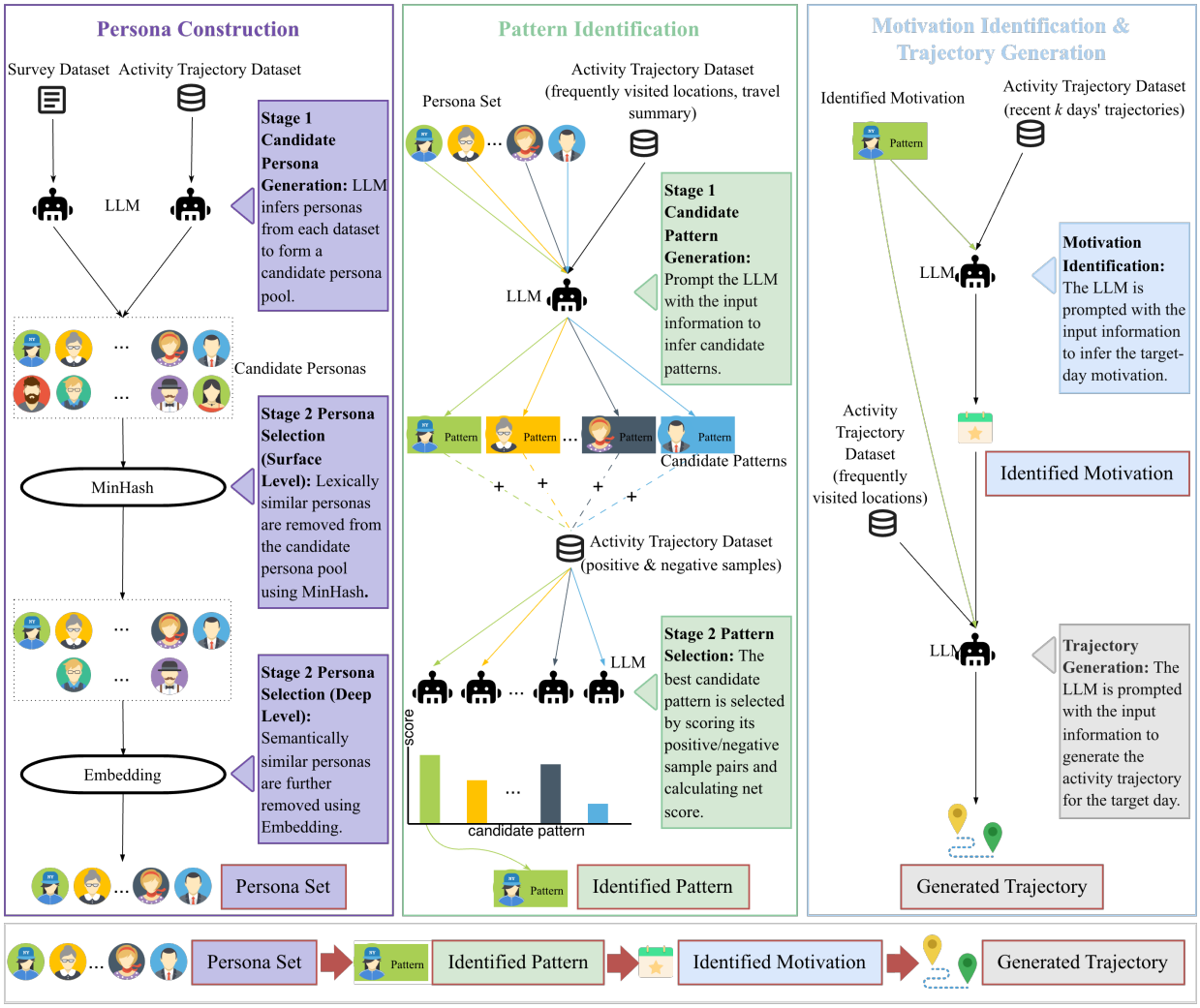


Figure 2: The overall framework of this study.

Building on this core mechanism, we introduce additional components to ensure that generated trajectories are realistic, diverse, and structurally complete. To enhance realism and diversity—where realism refers to consistency with real-world data and diversity reflects heterogeneous trajectories across individuals—we incorporate personas as semantic priors to guide the generation process

(Lutz et al. 2025). In addition, to enable structurally complete trajectory generation, we construct an enhanced travel summary that provides structured representations of both primary and secondary activities.

To operationalise this framework, we design a four-stage pipeline. First, an enhanced travel summary is constructed for each individual. Second, personas are derived using an LLM-based approach. Third, the LLM infers long-term habitual patterns. Fourth, the LLM infers short-term contextual motivations, which are then used to generate the activity trajectory for the target day. Details are provided in the following sections.

3.3. Enhanced Travel Summary

LLMs have limited ability to extract useful information from raw activity trajectories (X. Wang et al. 2023). We therefore construct an enhanced travel summary as a structured and aggregated representation of individual mobility behaviour, enabling LLMs to better understand it. Each travel summary consists of two parts, corresponding to weekday and weekend mobility. In each part, home and work activities are summarised by their timing, while secondary activities are summarised by both locations and timing. This design enables LLMs to capture the full structure of daily mobility patterns. The detailed information is presented in *Table 1*.

Table 1: Structure of the enhanced travel summary.

Day type	**Mobility aspect**	**Recorded information**
Weekday or Weekend	Home	• Typical first departure time from home • Typical final return time to home • Typical number and timing of returns to home between the first departure and the final return
	Work	• Typical number and timing of visits to work locations
	Secondary activities	• Typical location and timing of the first secondary activity visit • Typical location and timing of the last secondary activity visit • Typical locations and timings of the top-k most frequently visited secondary activities

3.4. Data-Driven Persona Construction

We adopt a data-driven persona construction pipeline (Ge et al. 2025), in which candidate personas are first generated and then filtered to obtain a compact and representative set. This approach avoids manual persona design and supports the generation of realistic and diverse trajectories.

Candidate personas are generated by providing each individual's mobility data to an LLM, which infers a corresponding persona. The input data are preprocessed into a structured JSON format to

facilitate model interpretation. We leverage two complementary datasets for persona generation. The first is the mobility dataset used for trajectory generation, which provides detailed individual-level travel records but lacks socioeconomic attributes. For this dataset, personas are inferred from observed mobility patterns, based on the assumption that individuals with different roles and lifestyles exhibit characteristic travel behaviours (Xu et al. 2018; Moro et al. 2021). The second is a temporally and geographically aligned travel survey dataset, which includes both mobility records and socioeconomic attributes, allowing the LLM to directly integrate behavioural and socioeconomic information to construct personas. The prompt templates used for persona generation are illustrated in *Figure 3* (trajectory-only dataset) and *Figure 4* (travel survey dataset).

```
Your task is to generate a persona based on the provided dataset.

You are given a JSON dataset derived from a mobile GPS dataset. It includes:
- A top-level "individual id";
- An "trajectories" array, where each element represents one day and contains that day's sequence of stay-point activities.

Think step by step:
- Read the person's daily trajectories and identify key mobility patterns;
- Based on these mobility patterns, infer plausible socioeconomic characteristics;
- Combine the inferred socioeconomic profile and travel behavior into a single persona.

Each persona must:
- Provide a concise description of this persona (maximum 30 words);
- Provide a role name based on the role description;
- Use professional, neutral language suitable for academic research;
- Follow this exact format:
[Role Name]: [Description of this role.]

Example:
Commuting Professional: Full-time employee, middle age, commuting daily by car, balancing family life with a stable income.
```

Figure 3: Example prompt template for persona generation based solely on mobility trajectories.

Because candidate persona generation produces one persona per individual record, the resulting persona pool is large and contains many redundant entries. We therefore apply a two-step persona selection strategy to remove highly similar personas and retain a compact and representative set for downstream trajectory generation. The first step removes near-duplicate personas based on lexical similarity using MinHash (Broder 1998), eliminating entries with identical or highly similar wording. The second step further filters semantically similar personas using embedding-based similarity, addressing cases where different expressions convey similar meanings. This two-step selection process ensures that the final persona set is both diverse and representative, and is subsequently used as semantic priors to guide trajectory generation.

3.5. Pattern Identification via Contrastive Evaluation

Pattern identification aims to infer individuals' long-term habitual mobility patterns. It is formulated as a process in which candidate patterns are first generated and the most representative pattern is selected for each individual.

In candidate pattern generation, the LLM is prompted to infer patterns conditioned on three inputs:

Your task is to generate a persona based on the provided dataset.

You are given a JSON dataset derived from the UK National Travel Survey. It includes:
- Top-level "individual id" and "household id";
- An "individual" object describing personal socioeconomic attributes;
- A "household" object describing household socioeconomic attributes;
- A "trip" array, where each element represents one day and contains the list of trips made on that day.

Think step by step:
- Read the person's socioeconomic attributes to form an initial understanding of their demographic and lifestyle background;
- Examine the person's daily trips to identify key travel behaviors;
- Combine the socioeconomic profile and travel behavior into a single persona.

Each persona must:
- Provide a concise description of this persona (maximum 30 words);
- Provide a role name based on the role description;
- Use professional, neutral language suitable for academic research;
- Follow this exact format:
[Role Name]: [Description of this role.]

Example:
Commuting Professional: Full-time employee, middle age, commuting daily by car, balancing family life with a stable income.

Figure 4: Example prompt template for persona generation using travel survey data.

the individual's travel summary, a persona sampled from the representative persona set, and additional frequently visited locations from the preprocessed data. Repeating this process across the persona set yields a set of candidate patterns $CP = \{cp_1, \ldots, cp_n\}$ for each individual, where each candidate pattern represents a persona-conditioned interpretation of the same mobility information. This design enables the candidate patterns to capture a diverse range of plausible habitual patterns for each individual. The example prompt template for candidate pattern generation is illustrated in *Figure 5*.

In pattern selection, we identify the most representative pattern using a self-consistency-based evaluation. The key idea is that a good pattern should align well with the individual's own mobility trajectories while remaining distinct from those of other users. To this end, each candidate pattern is evaluated against both positive samples (the individual's historical trajectories) and negative samples (trajectories sampled from other users), both obtained during data preprocessing. For each candidate pattern cp, we pair it with each trajectory sample and prompt the LLM to assign a rating r indicating how likely the individual would follow that trajectory under the given pattern. This produces one rating for each trajectory–pattern pair, resulting in a set of ratings for each candidate pattern. The overall score of a candidate pattern is then computed by aggregating these ratings across all samples:

$$\text{score}(cp) = \sum_{t \in T_i} r_t - \sum_{t' \in T_{\sim i}} r_{t'}. \tag{1}$$

Here, T_i denotes the set of positive samples for the individual, $T_{\sim i}$ denotes the set of negative samples, and r_t and $r_{t'}$ are the LLM ratings for individual trajectory–pattern pairs. In other words, this scoring mechanism rewards patterns that exhibit high consistency with the individual's own trajectories while penalizing those that align with other users' trajectories. This evaluation is applied to all candidate patterns, and the pattern with the highest score is selected as the individual's habitual

Your task is to generate a pattern description that reflects an individual's regular travel behavior based on the given contextual information.

You are provided with the following information:
- Persona: You are a !<INPUT 0>!. !<INPUT 1>!.
- Travel summary: !<INPUT 2>!.
- Additional frequently visited locations: !<INPUT 3>!.

Think step by step:
- Read the persona to understand the individual's background and lifestyle;
- Review the travel summary and the additional frequently visited locations to identify behavioral regularities;
- Based on these insights, describes the individual's typical mobility pattern.

Each output must:
- Remain realistic and internally consistent with the provided persona, travel summary, and locations;
- Consist of one coherent paragraph.

Text to Continue: I am !<INPUT 0>!. !<INPUT 1>!. My typical daily pattern...

Note:
!<INPUT 0>! persona name
!<INPUT 1>! persona description

Figure 5: Example prompt template for candidate pattern generation.

pattern. The example prompt template for pattern selection is illustrated in *Figure 6*.

3.6. Motivation Identification and Trajectory Generation

After identifying each individual's habitual pattern, we infer short-term motivation. Motivation inference follows a retrieval-based strategy that links an individual's motivation on the target day to their recent activity history (Park et al. 2023). This strategy is based on the intuition that recent activities reflect an individual's current behavioural trends, which in turn influence their subsequent mobility behaviour. Specifically, for a target date d, we retrieve the most recent consecutive k days of activity trajectories, or the closest available day if no such sequence exists. These trajectories, together with the individual's identified pattern, are provided to the LLM. The LLM then analyses recent behavioural trends and infers the individual's short-term motivation for the target day. The example prompt template for motivation inference is illustrated in *Figure 7*.

Finally, the identified long-term pattern and short-term motivation are jointly provided to the LLM to generate the individual's activity trajectory for the target day. The example prompt template for trajectory generation is illustrated in *Figure 8*.

3.7. Statement of Ethics

This study uses anonymised datasets in compliance with relevant data usage agreements and ethical requirements. The mobility dataset used for trajectory generation contains no personally identifiable information (PII) and was accessed under appropriate data-sharing agreements. The survey dataset was accessed under standard public data usage terms. All data were used exclusively for research purposes.

Your task is to evaluate how consistent a given daily trajectory is with a specified behavioral pattern.

You are provided with the following information:
- Behavioral pattern: !<INPUT 0>!
- Trajectory sample: !<INPUT 1>!

Think step by step:
- Read and understand the behavioral pattern to identify its core characteristic
- Examine the given trajectory and compare it with the behavioral pattern, focusing on the alignment of activities, temporal order, and spatial context.
- Assign a consistency rating from 1 to 10, and briefly explain your reasoning.

Each output must:
- Provide a numerical rating (1–10) representing the degree of consistency, where 10 = perfectly matches the behavioral pattern, 1 = completely inconsistent with the behavioral pattern.
- Include a concise explanation for the rating (1-2 sentences).
- Follow this exact format:
Rating: <fill in>. Reason: <fill in>

Example:
Rating: 8. Reason: The trajectory closely matches the commuting routine described in the pattern, including similar work hours and activity types. Minor deviations occur in evening leisure locations, but overall consistency is high.

Figure 6: Example prompt template for pattern selection.

Your task is to infer the individual's mobility motivation for today based on their established mobility pattern and recent trajectories.

You are provided with the following information:
- Identified mobility pattern: !<INPUT 0>!
- Recent trajectories: !<INPUT 1>!

Think step by step:
- Read the mobility pattern to understand the individual's long-term travel regularities and lifestyle;
- Analyze the recent trajectories to identify short-term deviations, new routines, or contextual factors;
- Integrate both long-term and short-term behavioral information to infer a plausible motivation for today's mobility.

Each output must:
- Provide a concise, realistic description (1–2 sentences) of the inferred motivation;
- Be consistent with both the behavioral pattern and recent trajectories.

Figure 7: Example prompt template for motivation inference.

Your task is to generate a realistic trajectory plan for today based on the given mobility pattern, current motivation, and frequently visited locations.

You are provided with the following information:
- Mobility pattern: !<INPUT 0>!
- Current motivation: !<INPUT 1>!
- Frequently visited locations: !<INPUT 3>!

Think step by step:
- Review the mobility pattern to understand the individual's long-term routines, anchor points, and mobility tendencies;
- Consider the current motivation and determine how it may influence today's schedule;
- Use the provided frequently visited locations to construct a plausible sequence of activities for today, including times and locations;
- Briefly explain the reasoning behind the generated trajectory.

Each output must:
- Be written strictly in the following JSON format;
{"plan": [<Location> at <Time>, <Location> at <Time>,...]
"reason":...}

Example:
{"plan": ["education#123 at 8:30", "eating and drinking#456 at 11:30", "education#789 at 12:50", "home#405 at 17:20"]
"reason":"My plan today includes attending school in the morning, taking a lunch break, returning to school in the afternoon, and going home afterward."}

Figure 8: Example prompt template for trajectory generation.

4. Experimental Setup

4.1. Data

We use two datasets in this study. First, we use an anonymised mobile in-app dataset derived from multiple applications, provided by Zhong et al. (2024). We focus on data from the Greater London area in October 2021 and retain users with at least 15 days of records, from which 100 individuals are randomly sampled for evaluation (*Figure 9*). The data are preprocessed to support LLM-based trajectory generation. Specifically, activity trajectories are extracted from the activity–trip chains in the provided dataset, where each activity is represented by its type and location identifier (*Table 2*). Each location identifier corresponds to a geographic coordinate in the dataset, which is retained for evaluation but not used as a model input. For each individual, the activity trajectories are then split into training and testing sets (8:2), and negative samples are constructed by randomly sampling trajectories from other users' training data. Each daily trajectory is further converted into a textual representation (e.g., "Activities on 2021/10/01: home#4144 at 00:21, ..."") to facilitate LLM-based reasoning. In addition, location visit frequencies are computed to identify frequently visited locations. To support persona construction, activity trajectories are also converted into a structured JSON format, as illustrated in *Figure 10*.

Table 2: Example of the activity trajectory dataset.

UserID	Start Time	End Time	Activity Type	Location ID
1	2021/10/01 00:21	2021/10/01 17:13	home	4144
1	2021/10/01 17:26	2021/10/01 17:27	others	63892011
1	2021/10/01 18:30	2021/10/01 18:36	work	202

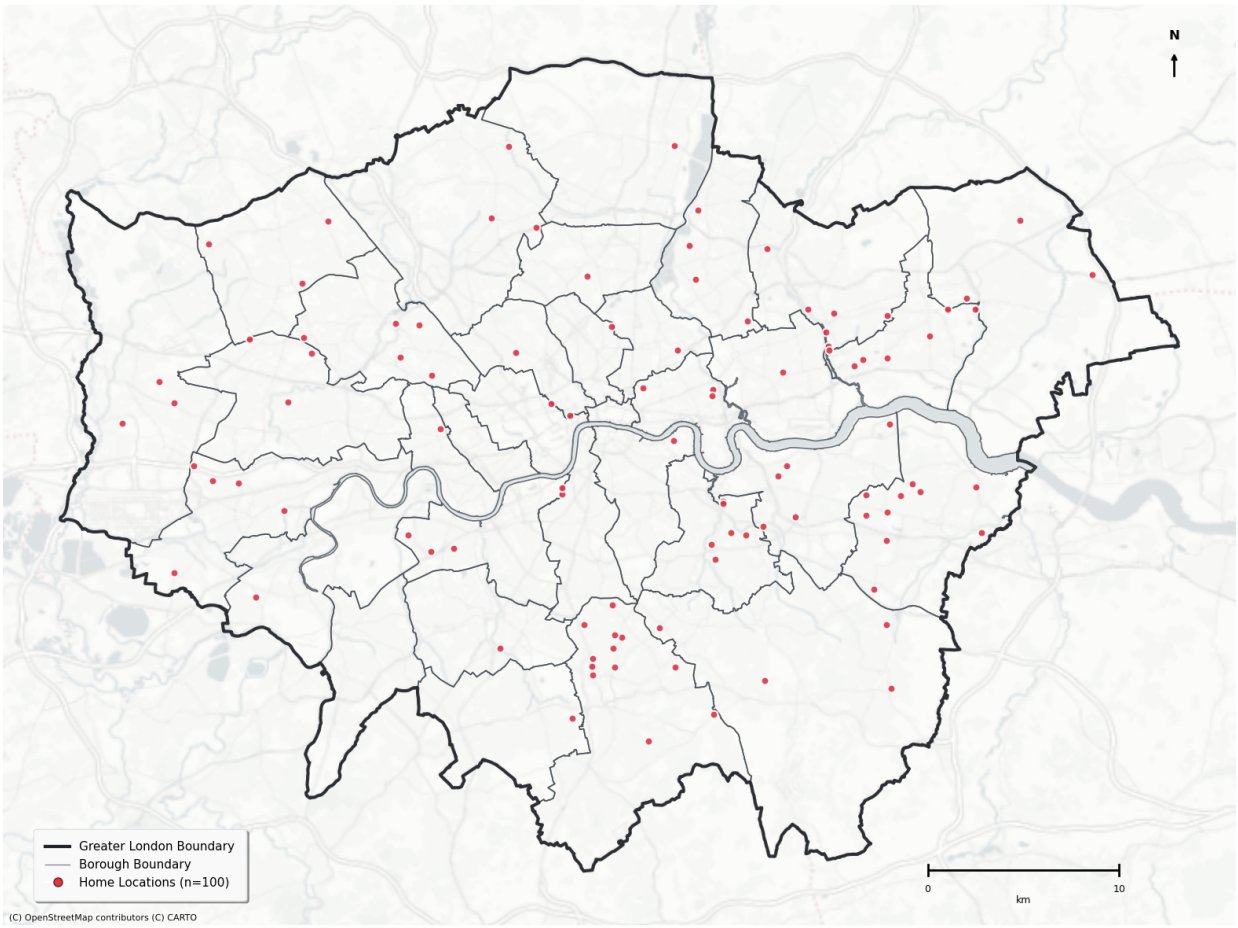


Figure 9: Spatial distribution of the sampled individuals' home locations within the Greater London Area.

```
{
 "individual id" : "1",
 "trajectories": [
  {
   "day": "2021-10-01, Friday",
   "stays": [
    {
     "stay sequence": 1,
     "location type": "home",
     "start time": "00:21",
     "end time": "17:13",
     "duration (minutes)": 1012
    },
    // ... remaining stays for this day (sequence 2-10)
   ]
  },
  // ... remaining trajectories for other days (10-02 to 10-24)
 ]
}
```

```
{
 "individual id": "2021000001",
 "household id": "2021000001",
 "individual": {
  "age band": "middle-aged adults",
  "working status": "Economically inactive: Permanent (retired, sick, disabled)",
  "bus usage frequency": "Never",
  "taxi usage frequency": "Never",
  "walking frequency": "Once or twice a month",
  "private car usage frequency": "At least once a day"
  // No records for this user: "workplace", "main transport to work", "main transport to school", "bicycle usage frequency".
 },
 "household": {
  "household income quintile": "1st"
 },
 "trip": [
  {
   "day": 1,
   "trips": [
    {
     "trip sequence": 1,
     "trip mode": "Car / van",
     "purpose of the origin": "Home",
     "purpose of the destination": "Food shopping",
     "start time": "11:00",
     "end time": "11:15",
     "trip distance (miles)": 3.0,
     "trip duration (minutes)": 15
    },
    // ... remaining trip for this day (sequence 2)
   ]
  },
  // ... remaining trips for other days (2-7)
 ]
}
```


Figure 10: Examples of JSON structures for mobile in-app data (left) and NTS data (right).

The UK National Travel Survey (NTS) is used alongside the mobile in-app data to support persona construction. The NTS provides mobility records with associated socioeconomic attributes, complementing the mobility-only information in the mobile in-app dataset. We use the 2021 NTS data to ensure temporal alignment with the first dataset. NTS data are preprocessed by converting individual records into a structured JSON format (Fig. 10).

4.2. Experimental Setup and Baselines

We implement the framework using OpenAI's GPT-4o-mini, as it offers an effective balance between performance and cost-efficiency. The model uses a temperature of 0.6–0.8 to avoid overly deterministic outputs at low temperatures and excessive randomness at high temperatures (P.-H. Wang et al. 2020; Wang, Liu, and Awadallah 2023).

In addition, in the persona selection step, we use a Jaccard threshold of 0.9 for lexical filtering and a cosine threshold of 0.55 for embedding-based semantic filtering using OpenAI's text-embedding-3-small. These thresholds are selected based on empirical tuning to effectively remove redundant personas while preserving sufficient diversity for downstream generation (Ge et al. 2025).

We evaluate our framework against three representative baselines:

- Markov: A probabilistic baseline. The state is the current activity (type#id) and temporal context (hour, day of week, and weekday/weekend). For each user, first-order transition probabilities and time distributions are estimated from the training data. Each test day starts from the observed first activity; the model then repeatedly samples the next activity and corresponding time, yielding a full-day activity trajectory.

- LSTM: A deep learning baseline. Each day is a fixed-length feature vector (activity counts, time statistics, and a normalised activity-type histogram). A separate two-layer model (32 and 16 units) is trained for each user using Adam and MSE. The model takes the previous three days' vectors and predicts the next day's vector, which is decoded into a full-day activity string. Each visit's activity type and location are then sampled uniformly from that user's training vocabulary.

- LLMob (J. Wang et al. 2024): A state-of-the-art LLM-based baseline. The model identifies a long-term pattern by prompting an LLM with ten manually defined roles (used in the original framework) and travel summaries (without explicitly treating home and work activities). Multiple candidate patterns are evaluated through LLM-based self-consistency scoring against historical routines, and the most consistent pattern is selected. Next, daily motivation is inferred from historical routines and, together with the selected pattern, used to generate each test day's full-day activity trajectory.

4.3. Evaluation Strategy

Model performance is evaluated by comparing the statistical (dis)similarity between real and generated data, using a metric applied to a set of mobility characteristics. The metric quantifies the (dis)similarity between real and synthetic data for each characteristic as a single numerical value

(Kapp, Hansmeyer, and Mihaljević 2024).

We use Jensen–Shannon Divergence (JSD) as the metric. It measures the difference between two probability distributions, defined as:

$$JSD(P\|Q) = \tfrac{1}{2}\mathbb{E}_P\left[\log\frac{P}{X}\right] + \tfrac{1}{2}\mathbb{E}_Q\left[\log\frac{Q}{X}\right] \tag{2}$$

where $X = \frac{P+Q}{2}$, and $\mathbb{E}_P$ and $\mathbb{E}_Q$ denote expectations with respect to distributions P and Q respectively (Kong et al. 2023). The value of JSD ranges from 0 to 1, where lower JSD values indicate better alignment between the generated and real data.

Four mobility characteristics are selected for evaluation from complementary perspectives.

- Step Distance (SD): Measures the spatial distance between consecutive activities, capturing movement ranges.
- Step Interval (SI): Measures the temporal gaps between consecutive activities, reflecting temporal dynamics.
- Daily Activity Routine Distribution (DARD): Captures the joint distribution of activity types and their occurrence times, reflecting activity type–time relationships.
- Spatial–Temporal Visit Distribution (STVD): Captures the joint distribution of activity locations and their occurrence times, reflecting spatiotemporal patterns

5. Results

5.1. Personas

We obtain a set of nine personas, as shown in *Table 3*. The persona set is compact yet diverse, capturing variations across life stages, socioeconomic conditions, and mobility styles. It spans children to older adults, reflects diverse employment and economic conditions, and covers a wide range of mobility patterns, from walking-dominated to car-reliant travel, as well as varied activity structures including commuting, home-centred, and multi-purpose travel.

5.2. Generative Performance

Table 4 compares the generative performance of the proposed method with baseline approaches. The proposed method achieves the best performance across all four metrics. Compared with LLM-Mob—the strongest baseline—our method reduces JSD by 77% for SD, 94% for SI, 65% for DARD, and 56% for STVD. These results indicate a substantially closer match to real-world mobility patterns.

Performance gains in SD and SI indicate improved modelling of spatiotemporal transitions between consecutive activities. These improvements can be attributed to the travel summary, which provides a structured "mobility skeleton" that guides the LLM toward generating trajectories with

Table 3: The nine personas constructed in this study.

Personas
Student Commuter: Young individual in the children and teenagers age band, traveling frequently for education, utilizing both car and walking as primary modes of transport.
Childhood Explorer: Active child or teenager, frequently walking and using a private car for social visits and personal errands, from a moderately affluent household.
Young Escort: A child or teenager from a lower-middle-income household, participating in regular short walking trips for escorting family members during personal errands.
Balanced Lifestyle Traveller: This persona strategically balances time between home, work, and leisure activities, with a focus on education and social engagement.
Active Young Adult: Unemployed young adult, cycling several times weekly for personal business, with a low household income.
Home-Based Routine Organizer: This individual maintains a structured daily routine with significant time spent at home, including varied activities such as shopping and work.
Structured Commuter: Middle-aged full-time employee, residing in a high-income household, commuting daily by car, consistently maintaining a structured work-life balance.
Part-Time Senior Walker: Senior individual, working part-time from home, with moderate income; engages in local walking trips and occasional car journeys for shopping and leisure.
Retired Car User: Retired individual, living in a low-income household, reliant on a private car for regular food shopping trips, with minimal use of public transport.

more realistic spatial ranges and temporal progression.

Improvement in DARD indicate that the proposed method better captures activity's type–time relationships. This improvement can be attributed, on the one hand, to the representation design that incorporates explicit activity type information, providing semantically meaningful inputs; and, on the other hand, to the LLM's semantic understanding and reasoning capabilities, which enable it to model plausible associations between activity type and time based on such inputs.

While the method also improves STVD, the gains are relatively smaller, which can be attributed to the limitation in the representation design: locations are represented using numerical identifiers (e.g., home#1, work#2) that do not encode geospatial relationships (e.g., distance, direction), limiting the LLM's ability to model geospatial relationships between locations.

Table 4: Comparison of generative performance between the proposed method and baseline approaches.

Models	**SD**↓	**SI**↓	**DARD**↓	**STVD**↓
Markov Model	0.0986	0.0078	0.1702	0.6802
LSTM	0.0988	0.1319	0.2729	0.6892
LLMob	0.0195	0.0356	0.1161	0.5297
Ours (w/o S)	0.0187	0.0277	0.1098	0.4702
Ours (w/o P)	<u>0.0052</u>	<u>0.0030</u>	<u>0.0418</u>	<u>0.2625</u>
Ours	**0.0045**	**0.0021**	**0.0405**	**0.2305**

5.3. Ablation Studies

We conduct ablation studies to assess the contribution of key components by comparing the full model ("Ours") with two variants, where each component is replaced by its baseline counterpart from J. Wang et al. (2024). Specifically, "Ours (w/o S)" uses their travel summary design, while "Ours (w/o P)" uses their persona construction approach. Results are shown in *Table 4*.

Removing the enhanced travel summary (w/o S) leads to substantial degradation across all metrics, with JSD increasing by 316% for SD, 1219% for SI, 171% for DARD, and 104% for STVD. This confirms the effectiveness of the enhanced travel summary.

Removing data-driven personas (w/o P) also degrades performance, though to a lesser extent, with increases of 16% for SD, 43% for SI, 3% for DARD, and 14% for STVD. This confirms the effectiveness of the data-driven approach for persona construction. The relatively smaller impact of personas compared to the travel summary arises because personas provide a coarser, high-level description of individuals' roles and behaviours, whereas the travel summary records individual-specific activity information, providing more informative and fine-grained signals for trajectory generation.

6. Conclusion

This study proposes an LLM-based framework for conditional individual-level daily activity trajectory generation that produces synthetic alternatives to raw individual mobility trajectories, reducing

reliance on directly observed data while preserving their analytical utility. The framework models daily activity trajectories as the interaction between long-term habitual patterns and short-term contextual motivations. To enhance realism, diversity, and structural completeness, we introduce two key components into the framework: a travel summary that serves as input information to provide structured mobility context for the model, and personas that act as semantic priors in prompts to guide diverse and realistic generation. Our innovations lie in the design of these components. First, we develop an enhanced travel summary that enables structurally complete trajectory generation. Second, we propose a data-driven persona construction pipeline that enables reproducible persona generation and facilitates its application across datasets. Experimental results show that the proposed framework consistently outperforms existing approaches, while ablation studies further demonstrate the effectiveness of the two innovations.

Despite these contributions, several limitations remain. First, activities are represented using activity type and ID labels (e.g., home#1). This representation design provides explicit information about activity types, but the numerical IDs do not explicitly encode geospatial information. As a result, the model is limited in modelling activity locations. Future work could incorporate geospatial representations to improve location modelling. Second, the framework is evaluated on a single region and time period, and future work should examine its generalisability across diverse contexts. Finally, formal evaluation of privacy guarantees and validation on downstream tasks are not explored in this study.

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